

CRAMPS — Experimental Supplement

Full Documentation of Experiments, Methods, and Results

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Abstract

This supplement documents every experiment supporting the CRAMPS paper: the exact model definition, all six experimental procedures, their parameters, their results, and the reproducibility instructions. It is intended as the companion artifact for the code repository and as a reference for reviewers verifying the claims. Each section corresponds to a numbered experiment with its hypothesis, method, parameters, raw results, and interpretation, including the honest statistical caveats that apply.

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1 The Model Under Test

Before the experiments, we fix the exact system. CRAMPS is defined by four components operating in a loop over discrete epochs.

1.1 Graph

A weighted undirected graph $G = (V, E, W, \Phi)$ where each node v_i carries a semantic embedding $\phi_i \in \mathbb{R}^d$. Edges exist where embedding cosine similarity exceeds θ_{sim} ; weights initialize from that similarity.

1.2 Agent scoring function

Each agent a scores candidate next-node j from current node i by:

$$f_a(j \mid i, G) = \alpha_a \text{sem}(i, j) + \beta_a \text{attr}(j, G) + \gamma_a \text{need}(j, g_a) + \delta_a H(j, G) \quad (1)$$

where sem is local cosine similarity, attr is PageRank (structural attraction), need is cosine proximity to the agent’s goal vector g_a , and H is neighborhood weight entropy. The tuple $(\alpha, \beta, \gamma, \delta)$ is the agent’s *scoring policy*; T_a is its softmax temperature.

1.3 Competitive Hebbian plasticity

After an agent walks trajectory τ_a , each traversed edge is reinforced proportionally to the path’s *coherence* $r(\tau_a)$ (mean cosine similarity of path embeddings to the path centroid):

$$w_{ij} \leftarrow \min(w_{ij} + \eta r(\tau_a) \cos(\phi_i, \phi_j), W_{\max}) \quad (2)$$

1.4 Selective decay

Every epoch, all edges decay multiplicatively and weak edges are pruned:

$$w_{ij} \leftarrow w_{ij}(1 - \lambda), \quad \text{remove } (i, j) \text{ if } w_{ij} < \theta_{\text{prune}} \quad (3)$$

1.5 Fixed parameters across all experiments

Parameter	Symbol	Value
Learning rate	η	0.05
Max edge weight	$W_{\max}$	3.0
Prune threshold	θ_{prune}	0.005–0.007
Walk length	L	8 steps
Top- k neighborhood	k	6–10
Embedding dim (synthetic)	d	32
Embedding dim (narrative)	d	384 (all-MiniLM-L6-v2)
Graph model	—	Watts-Strogatz ($k = 4, p = 0.2$)

Table 1: Parameters held constant across all six experiments.

2 Experiment 1 — Static Strategy Comparison

Hypothesis. Different scoring policies produce measurably different navigation behavior (loop formation, topic coherence, diversity) on a fixed graph.

Method. Build a single graph from the narrative corpus. Run three agent archetypes (Conformist, Disruptive, Opportunist) as walkers *without* plasticity. Measure per-agent: path coherence, node revisit rate (loops), and unique-node diversity.

Parameters. $N \approx 78$ nodes, 5 runs per agent, walk length 8.

Result. The archetypes separate cleanly: the Conformist produces high-coherence looping paths around central nodes; the Disruptive produces low-coherence dispersed paths; the Opportunist is intermediate. This validates that the scoring function meaningfully differentiates behavior before any plasticity is introduced.

Role in paper. Establishes that the three archetypes are behaviorally distinct. Foundational; not a headline claim.

Script. `eval/run_experiments.py`

3 Experiment 2 — Plasticity Dynamics

Hypothesis. When walks reinforce edges (Hebbian update) and edges decay, structural inequality (Gini) evolves over epochs toward a stable fixed point.

Method. Single narrative graph, single agent ecology, full plasticity loop. Track Gini per epoch over 80 epochs at low decay ($\lambda = 0.01$).

Result. Gini rises from 0.098 to 0.196 and stabilizes ($\Delta_{\text{epochs } 61-80} = -0.00025$). The system reaches a stable structural fixed point rather than diverging or oscillating.

Role in paper. Section 4.4 (convergence). Shows the dynamics are well-behaved.

Script. `eval/run_plasticity.py`

4 Experiment 3 — Phase Diagram (Decay Sweep)

Hypothesis. Structural inequality depends non-monotonically on the decay rate λ ; there exists a critical λ that maximizes inequality.

Method. *Neutral synthetic graphs* (Watts-Strogatz, $N = 20$, random embeddings, no corpus bias) to isolate the decay effect from any semantic structure. Two *pure* agents compete: a pure attractor ($\beta = 1$, all else 0) and a pure explorer ($\delta = 1$, all else 0). Sweep $\lambda \in \{0.000, 0.025, \dots, 0.500\}$ (21 values), 15 seeds each — 315 total runs of 80 epochs.

Result. A sharp non-monotonic curve. This is the paper’s most robust result.

λ	Gini (mean)	Attractor dominance
0.000	0.126	73%
0.100	0.243	0%
0.200	0.310	22%
0.300	0.387	18%
0.400	0.280	10%
0.500	0.000	0%

Table 2: Phase analysis with pure agents (315 runs). Peak Gini = 0.387 at $\lambda \approx 0.3$. Note: this 0.387 is the *pure-agent* peak, distinct from the heterogeneous-archetype peak (0.360) in Experiment 5.

Three regimes.

$\lambda \rightarrow 0$ (**memory preservation**): all traces persist, Gini moderate (0.126).

$\lambda \approx 0.3$ (**selective consolidation**): decay prunes weak paths but not dominant ones; Gini maximized (0.387).

$\lambda \geq 0.5$ (**critical dissipation**): structure destroyed faster than built; Gini collapses to 0.

Role in paper. Proposition 1, Table 2, Figure 1A. The central robust finding.

Script. `eval/run_phase_analysis.py`

5 Experiment 4 — Counterfactual Head-Start

Hypothesis. If dominance were driven by first-mover advantage, giving the weaker (Disruptive) agent exclusive early access should reduce the final gap.

Method. Three conditions on the narrative corpus ($n \approx 78$): simultaneous start ($K = 0$), Disruptive head-start of $K = 10$, and $K = 50$ epochs. After the head-start, all agents run jointly to epoch 80. Measure final goal proximity per agent.

Result. Dominance is *not* reduced by head-start — it *widens*:

Condition	Conformist	Disruptive	Opportunist	$\Delta(C-D)$
Simultaneous ($K = 0$)	0.547	0.193	0.397	0.354
Disruptive HS ($K = 10$)	0.548	0.183	0.394	0.365
Disruptive HS ($K = 50$)	0.544	0.175	0.391	0.369

Table 3: Counterfactual: head-start widens the gap ($0.354 \rightarrow 0.369$, +4.2%).

Honest caveat. This is a +4.2% effect on $n = 40$ texts without confidence intervals. It is suggestive, not conclusive. The “paradox” label should be read as a hypothesis requiring higher-powered replication.

Role in paper. Propositions 2 and 3, Figure 1B.

Script. `eval/run_counterfactual.py`

6 Experiment 5 — The Kill Test (Policy Homogeneity)

This is the adversarial experiment designed to falsify the paper’s original hypothesis. It is the most important methodological move in the project.

Hypothesis under attack. The original claim was: “scoring policy asymmetry ($W_{\text{ATTRACTOR}}$) causes topological capture.” If true, then removing the asymmetry (giving all agents identical scoring) should eliminate or sharply reduce the inequality.

Method. Two conditions, both at $N = 20$, 80 epochs, 12 seeds, full λ sweep:

- **Condition A (heterogeneous):** three archetypes with different scoring weights (Conformist, Disruptive, Opportunist).
- **Condition B (homogeneous / kill):** all three agents share *identical* scoring weights ($\alpha = 1.0, \beta = 0.7, \gamma = 0.6, \delta = -0.1, T = 1.1$), differing only in goal position (centroid, peripheral, dominant).

If the original hypothesis holds, Condition B should show much lower Gini.

Result. It does not. The peaks are nearly equal but at different λ :

λ	Gini-A (heterog.)	Gini-B (homog.)	Δ
0.00	0.158	0.159	−0.001
0.10	0.328	0.290	+0.038
0.20	0.327	0.330	−0.003
0.25	0.346	0.359	−0.012
0.30	0.360	0.333	+0.027
0.40	0.252	0.259	−0.007
0.50	0.234	0.224	+0.011

Table 4: Kill test. Peak-to-peak Δ Gini = 0.001 (A peaks at 0.30, B at 0.25).

Interpretation — with the honest statistical caveat. The original hypothesis is *not supported*: removing scoring asymmetry did not remove the inequality. However, this is a **null result under limited statistical power**: with 12 seeds, no confidence intervals, peaks at different λ , and point-wise differences ranging −0.012 to +0.038, we can only say *we do not detect a difference*, not that the conditions are equivalent. Absence of evidence is not evidence of absence.

What survives. What the kill test legitimately establishes: scoring asymmetry is *not necessary* for the phase diagram to appear. The $\lambda \approx 0.3$ peak is present in both conditions. This redirects the contribution away from “policy asymmetry causes capture” toward “goal-position heterogeneity under selective forgetting suffices.”

Known circularity. In Condition A, the Conformist’s goal is the corpus centroid and the reward is coherence (closeness to centroid). This is a partial tautology: the agent aligned with the reward reference scores higher by construction. The kill test exposes this rather than fully resolving it.

Role in paper. Proposition 4, Table 5. Reframes the entire contribution.

Script. `eval/run_kill_test.py`

7 Experiment 6 — Reward Function Transitions (Preliminary)

Status: preliminary. This experiment is *not* a formal claim in the paper. It is documented here as the basis for future work.

Hypothesis. The dominant agent is determined by *which reward the system uses*, not by agent identity. Changing the reward should change who wins.

Method. Condition B (homogeneous scoring) at $\lambda = 0.3$, 12 seeds, swapping the reward function: coherence, novelty, path-diversity, anti-centrality.

Reward	Gini	Dominant agent
Coherence	0.333	centroid
Novelty	0.355	peripheral
Diversity	0.275	centroid
Anti-central	0.345	peripheral

Table 5: Reward function determines the winner. Preliminary, 12 seeds, single λ .

Result.

Interpretation. Under novelty and anti-centrality rewards, the peripheral agent dominates — the centroid is not inherently hegemonic. The diversity anomaly (centroid still wins, but lowest Gini) is hypothesized to arise because the centroid has more available neighbors, enabling longer unique-node paths. This requires a full λ sweep with ≥ 20 seeds before becoming a formal claim.

Role in paper. Future work only. Mentioned in Limitations.

Script. `eval/run_kill_test.py` (reward variants)

8 Summary of Evidence Strength

Claim	Evidence	Strength
Non-monotonic λ phase diagram	315 runs, synthetic	Strong
Peak at $\lambda \approx 0.3$	315 runs, synthetic	Strong
Stable Gini fixed point	80 epochs, narrative	Solid
Kill test (policy not necessary)	12 seeds	<i>Null, low power</i>
Counterfactual head-start widens gap	$n = 40$, no CIs	<i>Suggestive</i>
Reward determines winner	12 seeds, 1 λ	<i>Preliminary</i>

Table 6: Honest assessment of each claim’s evidential strength.

9 Reproducibility

```
Python 3.12
networkx==3.6
sentence-transformers
scipy==1.17
numpy==2.4
matplotlib
```

```
# Phase diagram (Experiment 3)
python eval/run_phase_analysis.py

# Counterfactual (Experiment 4)
python eval/run_counterfactual.py

# Kill test + reward variants (Experiments 5, 6)
python eval/run_kill_test.py

# Full pipeline
python eval/run_full.py
```

Seeds. All experiments use fixed seeds (base seed 42) for exact reproducibility. The synthetic graph experiments seed both NumPy and Python `random`.

Determinism note. The narrative-corpus experiments depend on the `all-MiniLM-L6-v2` embedding model; results are deterministic given the same model version and the same filtered corpus (289 raw $\rightarrow$ 76–78 filtered, seed 42).